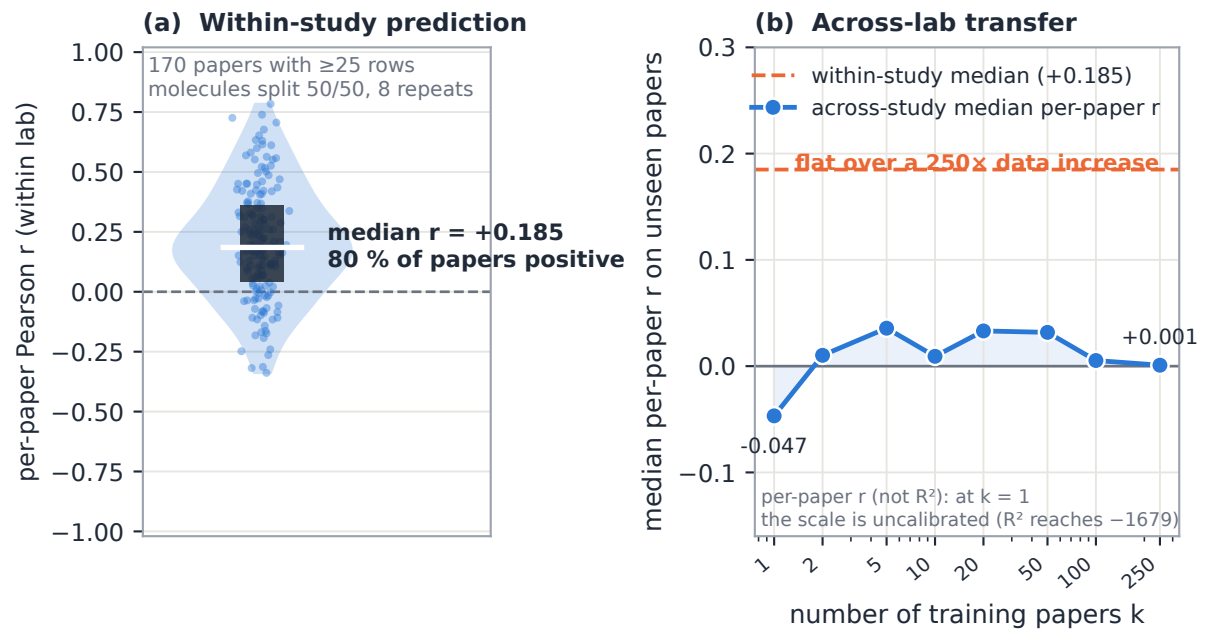

(a) Within-study prediction
170 papers with ≥25 rows
molecules split 50/50, 8 repeats
median r = +0.185
80 % of papers positive
per-paper Pearson r (within lab)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
(b) Across-lab transfer
within-study median (+0.185)
across-study median per-paper r
flat over a 250× data increase
+0.001
-0.047
per-paper r (not R²): at k = 1
the scale is uncalibrated (R² reaches −1679)
median per-paper r on unseen papers
0.3
0.2
0.1
0.0
−0.1
1
2
5
10
20
50
100
250
number of training papers k